Highlights

UAV-TAlign-12K: Benchmarking Scene-Level Reliability in Cross-FOV UAV Infrared-Visible Alignment

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- UAV-TAlign-12K provides 6,039 real cross-FOV UAV infrared-visible pairs.
- Seven baselines establish broad pairwise homography availability.
- UAV-TAlign produces 15/15 scene transforms and retains 9 at the strict gate.
- Resampling and controlled perturbations validate the reliability protocol.

UAV-TAlign-12K: Benchmarking Scene-Level Reliability in Cross-FOV UAV Infrared-Visible Alignment

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Abstract

Reliable UAV infrared-visible alignment is important for aerial inspection, nighttime monitoring, and multimodal thermal sensing, yet existing resources provide limited support for evaluating real captured cross-FOV alignment at the scene level. We introduce UAV-TAlign-12K, which contains 6,039 real-world UAV infrared-visible pairs across 15 scenes and provides an integrity-checked 6,037-pair official evaluation split. The dataset spans day, night, and low-light capture, grayscale and pseudocolor thermal rendering, and wide, zoom, and standard UAV views. We further define a manifest-fixed, label-free benchmark that separates pairwise homography evidence from scene-level retained alignment and reports condition-aware and risk–coverage diagnostics. Evaluation with seven off-the-shelf baselines, spanning classical, infrared-visible-specific, and modern learned methods, shows that pairwise evidence is broadly available: RIFT2, raw MINIMA, and RoMa produce homographies for all 6,037 evaluated pairs, while LoFTR and XoFTR remain near-saturated. As the benchmark reference framework, UAV-TAlign aggregates frame evidence through reliability-guided progressive selection, robust consensus, and QA-aware verification. It produces scene homographies for all 15 scenes and retains 9 under the strict canonical gate. Five-fold leave-frame-out analysis yields a 1.39-pixel median scene-consensus drift, while controlled translation, rotation, and scale perturbations produce monotonic degradation of the joint QA signal. Together, the dataset, protocol, and reference framework establish a reproducible benchmark for studying when cross-FOV UAV infrared-visible alignment is reliable enough to retain.

Keywords: infrared-visible dataset, UAV imagery, multimodal registration, alignment benchmark, scene-level reliability

1. Introduction

Infrared-visible alignment is becoming a foundational capability for UAV inspection, monitoring, and nighttime multimodal perception. Visible imagery provides semantic and structural detail, while infrared thermal imagery remains informative under illumination changes, smoke, haze, and low-light conditions. This complementarity has driven progress across multispectral pedestrian perception, thermal object detection, visible-thermal tracking, and UAV multimodal sensing [1, 2, 3, 4, 5, 6, 7, 8]. For these systems to operate as fused sensors rather than separate streams, however, RGB and thermal observations must be aligned reliably at the scene level.

Real UAV infrared-visible alignment is substantially more demanding than standard same-view image matching. Aerial platforms introduce viewpoint drift, cross-FOV capture, rolling scale changes, and cross-resolution sensors; thermal images further introduce low-texture regions, grayscale or pseudocolor rendering, and strong day/night appearance shifts. These factors make individual frame-level homographies uneven: some pairs are strongly supported, while others require additional scene context before they should influence the final transform. The practical question is therefore not only whether a matcher can produce correspondences for one pair, but whether a system can decide which scene-level alignment should be trusted.

Recent pairwise matching has advanced rapidly. Transformer, dense, and multimodal matchers such as LoFTR, LightGlue, DKM, RoMa, XoFTR, and MINIMA have substantially improved correspondence availability under difficult appearance changes [9, 10, 11, 12, 13, 14, 15, 16]. At the benchmark level, recent UAV visible-infrared alignment studies have also shown that synthetic transform supervision can support trainable alignment baselines with controlled geometric metrics [17]. UAV-TAlign-12K targets the complementary operating regime: real captured cross-FOV UAV infrared-visible scenes where the central product is a retained scene-level transform rather than only a supervised pairwise transform prediction. Our 12K-scale experiments confirm this progress in the UAV infrared-visible setting: strong off-the-shelf classical and modern baselines already provide broadly available pairwise homography evidence on UAV-TAlign-12K. This result creates a concrete tension: a method can approach saturated pairwise usability while producing only selective scene-level retention under a strict acceptance rule.

Pairwise usability and scene-level canonical acceptance therefore measure different operating questions and should not be read as the same success criterion. The key step is to aggregate evidence across a scene, suppress unstable homographies, and verify whether the final transform should be retained.

Pairwise homographies can be available for most frames while their support and scene consistency remain uneven. Robust consensus and QA therefore determine whether the resulting scene transform is retained, whereas the reliability score ranks scenes only for operating-profile analysis.

This observation motivates a benchmark-first framing with two complementary evaluation questions: whether a method can produce usable pairwise homography evidence, and whether multi-frame evidence supports a scene transform that should be retained. Treating these as separate units prevents high pairwise availability from being interpreted as automatic scene-level reliability.

Our primary contribution is UAV-TAlign-12K, a real-world UAV infrared-visible alignment dataset with 6,039 candidate pairs and an integrity-checked 6,037-pair evaluation split across 15 scenes. The collection spans day, night, low-light, grayscale thermal, pseudocolor thermal, and wide, zoom, and standard UAV views. Its manifest-fixed, label-free protocol evaluates pairwise homography evidence, canonical scene retention, condition-aware reliability, and risk–coverage operating profiles. Together, the data and protocol make scene-level alignment reliability directly studyable under captured cross-FOV UAV conditions.

To instantiate the scene-level benchmark track, we present UAV-TAlign as its reference alignment framework. It operates on top of strong pairwise matchers and uses reliability-guided progressive evidence selection, robust scene consensus, and QA-aware verification to decide whether a scene-level transform should be retained. The framework provides a reproducible implementation of the reliability question exposed by UAV-TAlign-12K.

Our contributions are threefold:

- We introduce UAV-TAlign-12K, a 6,039-pair real captured cross-FOV UAV infrared-visible dataset with an integrity-checked 6,037-pair official split across 15 diverse scenes.

- We establish a manifest-fixed benchmark that separates pairwise homography evidence from canonical scene-level retention and supports condition-aware and risk–coverage analysis.

- We provide UAV-TAlign as the scene-level benchmark reference framework and use classical, infrared-visible-specific, and modern off-the-shelf baselines, cumulative ablations, and controlled protocol validation to demonstrate its reliability-centered evaluation capability.

2. Related Work

2.1. Classical Image Registration

Classical image registration pipelines typically rely on hand-crafted local features followed by robust geometric fitting. Representative methods built on SIFT, SURF, ORB, BRISK, KAZE, or AKAZE remain attractive because they are simple, interpretable, and do not require task-specific training [18, 19, 20, 21, 22, 23]. In cross-modal settings, the appearance gap between visible and thermal imagery can reduce descriptor repeatability, especially when thermal frames contain low-texture or homogeneous regions. These methods nevertheless provide a strong geometric reference point for understanding which parts of the registration problem are addressed by local feature extraction and which parts require higher-level scene reasoning.

These methods remain important baselines in our study because they represent the classical registration view of the problem: stable local features followed by robust homography fitting. Robust model estimation remains central in these pipelines, from RANSAC and multiple-view geometry to more recent variants such as MAGSAC and MAGSAC++ [24, 25, 26, 27]. Our results show that real UAV infrared-visible scenes benefit from moving beyond a single-pair formulation toward scene-level evidence aggregation and reliability control.

2.2. Deep Pairwise Matching for Multimodal Registration

Recent dense and semi-dense matching methods have significantly improved pairwise geometric correspondence estimation. Learned local features have evolved from trainable detectors and descriptors such as SuperPoint, LF-Net, D2-Net, R2D2, DISK, S2DNet, HardNet, L2-Net, SOSNet, GeoDesc, ContextDesc, ASLFeat, ALIKE, and Key.Net [28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41], while match filtering and end-to-end correspondence reasoning have progressed through SuperGlue, NCNet, Patch2Pix, COTR, LoFTR, ASpanFormer, LightGlue, DKM, RoMa, Efficient LoFTR, OmniGlue, MINIMA, and XoFTR [9, 42, 43, 44, 10, 11, 12, 13, 14, 45, 46, 16, 15]. In particular, these methods are better at recovering useful correspondences in low-texture regions, repetitive structures, or substantial appearance shifts. Broad reviews of this transition from classical pipelines to trainable matching, together with practical benchmarking studies, provide additional context [47, 48].

Our work builds directly on this progress by treating modern matchers as high-capacity evidence generators. The empirical picture in UAV-TAlign-12K is that strong off-the-shelf multimodal matchers and infrared-visible classical baselines already provide very high pairwise availability. This moves the central question from correspondence generation alone to the conversion of frame-level estimates into stable and reliable scene-level alignment under realistic UAV operating conditions.

2.3. Visible-Thermal Registration and Evaluation

Visible-thermal registration has been studied under a range of assumptions, from same-view or near-same-view alignment to broader multimodal correspondence learning. Radiation-variation insensitive matching methods such as RIFT and deep alignment or stitching formulations such as CLKN and UDIS indicate the breadth of existing registration formulations [49, 50, 51]. Many existing settings emphasize relatively controlled viewpoint changes or pairwise registration outcomes. By contrast, practical UAV infrared-visible data frequently combine cross-FOV capture, cross-resolution sensing, thermal rendering variation, and low-light conditions. These factors make scene-level reliability a more central concern than in many standard visible-thermal evaluation setups.

Recent UAV registration resources now span several complementary evaluation regimes. UAVMatch provides synthetic transform supervision and a trainable alignment baseline with controlled geometric metrics [17]. The 7,969-triplet ATR-UMMIM benchmark provides real UAV visible–infrared data, a semi-automated pixel-level reference, and condition attributes [52], while UAV-TIRVis provides 80 manually registered aerial thermal-visible pairs for direct image-domain evaluation [53]. UAV-TAlign-12K targets a distinct multi-frame operating question: it fixes an integrity-checked captured cross-FOV split, evaluates heterogeneous off-the-shelf evidence generators, and studies how frame-level homographies become retained scene products under a scalable reliability protocol.

UAV-TAlign-12K extends this evaluation landscape by packaging real cross-FOV infrared-visible scenes together with scalable label-free diagnostics. The protocol measures scene consistency and relative alignment quality, allowing the study to focus on reliability-controlled scene-level alignment under practical UAV acquisition conditions rather than isolated pairwise correspondence counts.

2.4. UAV Multimodal Datasets and the Remaining Gap

UAV multimodal datasets have supported progress in detection, tracking, segmentation, and broader cross-modal perception, spanning aligned pedestrian and automotive benchmarks as well as visible-thermal detection and tracking benchmarks such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV [1, 2, 3, 4, 5, 6, 7, 8]. These resources have primarily advanced downstream perception tasks, while scene-level infrared-visible alignment places additional emphasis on reliability-aware geometric decision making. In particular, a practical registration pipeline must decide not only how to align frames, but also when the scene-level evidence is strong enough to trust the resulting transform.

This is the gap our paper targets. Complementary to pairwise geometric-accuracy benchmarks and new matcher architectures, our primary focus is a real captured dataset and benchmark protocol for cross-FOV UAV infrared-visible alignment. The collection and label-free protocol make pairwise evidence, retained scene transforms, and condition-dependent reliability directly studyable. UAV-TAlign instantiates this scene-level benchmark as a reproducible reference framework.

3. UAV-TAlign-12K and Evaluation Protocol

3.1. Dataset Scope

Our primary contribution is UAV-TAlign-12K, a dataset and benchmark for real UAV infrared-visible registration. The collection contains 6,039 candidate RGB-infrared pairs, corresponding to 12,078 images across 15 UAV scenes. The official evaluation split contains 6,037 integrity-checked RGB-infrared pairs and 12,074 images. Each sample is organized as an RGB image paired with a thermal image captured from the same scene context, while the collection spans practical UAV conditions that go beyond standard near-same-view multimodal matching. In scope and modality, it is closer to recent visible-thermal benchmarks such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV than to classical same-sensor image matching setups [1, 2, 3, 4, 5, 6, 7, 8].

Table 1: Positioning of UAV-TAlign-12K relative to representative infrared-visible and UAV multimodal resources. UAVMatch denotes the recent synthetic transform-supervised UAV visible-infrared alignment benchmark.

Resource	Data basis	Alignment signal	Scale	Primary evaluation focus
KAIST / FLIR / LLVIP / M3FD	Ground or mixed RGB-infrared data	Paired or aligned imagery	Dataset-specific	Detection, fusion, low-light perception
DroneVehicle / VTUAV / LasHeR / Anti-UAV	UAV or tracking-centric RGB-infrared data	Task annotations	Dataset-specific	Detection or tracking under multimodal sensing
UAVMatch / TGRS [17]	Synthetic transforms from UAV multimodal data	Affine / homography supervision	81K train / 15K test pairs	Supervised transform regression and geometric metrics
ATR-UMMIM [52]	Real UAV visible / infrared / registered-visible triplets	Semi-automated pixel-level reference	7,969 triplets	Pairwise registration accuracy and condition attributes
UAV-TIRVis [53]	Real UAV thermal-visible imagery	Manual pairwise registration	80 pairs	Direct image-domain registration quality
UAV-TAlign-12K	Real captured cross-FOV UAV infrared-visible data	Manifest-fixed scene reliability protocol	6,039 candidate / 6,037 valid pairs	Pairwise evidence and scene-level retained alignment

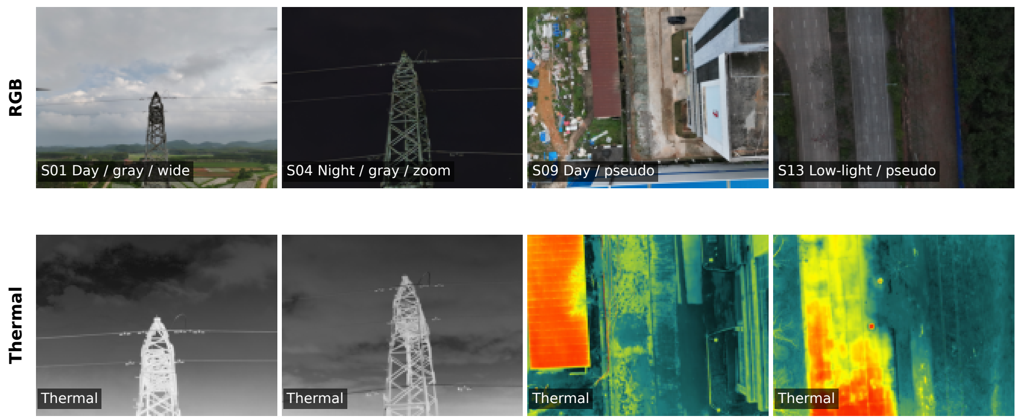

Figure 1: Representative UAV-TAlign-12K RGB-infrared pairs spanning day, night, low-light, grayscale-thermal, pseudocolor-thermal, wide-view, and zoom-view conditions.

All main journal-scale experiments are tied to the official manifest with canonical SHA256 prefix `a092f7ad...bb202c6c`. This manifest-based entry point fixes the evaluated split and prevents direct folder traversal from silently changing the reported results. We retain UAV-TAlign-1K-Lite as a development and ablation subset, while UAV-TAlign-12K is the main benchmark-scale evaluation collection for this article.

At the scene level, UAV-TAlign-12K preserves the factors that make reliability assessment meaningful: day, night, and low-light capture; grayscale and pseudocolor thermal rendering; and both wide-view and zoom-view scene families. The packaged formal split includes 7 day scenes, 5 night scenes, and 3 low-light scenes; 8 grayscale-thermal scenes and 7 pseudocolor-thermal scenes; and paired wide/zoom controls for the substation scene family.

3.2. Acquisition and Curation

UAV-TAlign-12K was acquired between the second half of 2024 and the first half of 2026 across multiple industrial and natural scenes in southern

China. All 15 scenes were captured with the same DJI Matrice 350 RTK platform and Zenmuse H30T multisensor payload. The integrated payload automatically synchronized its visible and infrared products, so no manual geometric alignment was used to construct the released RGB-infrared pairs. The flights combined hovering and route-based capture. All operations used a registered aircraft, a CAAC-qualified pilot, and permitted flight areas.

The released streams retain the native H30T acquisition modes. Wide-camera stills are 4032×3024, zoom-camera stills are 3664×2748, and thermal images are 1280×1024. In scenes 01 and 02, a short 3840×2160 segment was produced by triggering still capture while H30T video recording was active; these images are not extracted video frames. Scene 07 is the dedicated video-derived sequence: synchronized RGB and infrared streams were sampled at 2 frames per second over one complete orbit around a transmission tower. The grayscale scenes use the H30T *White Hot* setting, whereas all pseudocolor scenes use the on-device *Hot Iron* palette.

Dataset curation was performed before algorithm evaluation. Images were removed when they did not capture the intended scene subject or showed clear acquisition-quality problems, including abnormal exposure, inaccurate focus, or damaged data. No image was selected or removed according to the output of any matcher, alignment method, or QA score. The resulting collection contains 6,039 candidate synchronized pairs; the official split subsequently excludes only the two thermal files that do not pass the documented integrity check.

Release preparation removes embedded metadata and normalizes filenames while preserving image pixels. RGB and infrared files are paired through their shared H30T capture key, ordered by capture time and sequence index, and renamed with a common six-digit pair identifier. No resizing, cropping, rotation, recompression, or image enhancement is applied. The laboratory owns the collection, and the release package is prepared under the Creative Commons Attribution 4.0 International license.

3.3. Dataset Organization

Each scene is stored under its own scene directory, and paired samples are formed by matching file stems between `rgb/` and `thermal/` subdirectories. The dataset loader constructs scene pairs from common RGB and thermal filenames, while scene-level metadata are loaded from the official manifest. For each pair, the loader stores scene identity, pair identity, light condition, thermal rendering, view type, and scene label.

This organization supports both pairwise baseline evaluation and scene-level pipeline evaluation under the same manifest definition, while also enabling condition-based breakdowns from a unified scene manifest.

3.4. Complementary Evaluation Units

The protocol uses two complementary evaluation units. First, off-the-shelf baselines are evaluated at the pair level. Each of the 6,037 integrity-checked RGB-infrared pairs is processed independently, and the resulting statistics describe whether a method can produce usable pairwise homography evidence under a fixed off-the-shelf setting.

Second, UAV-TAlign is evaluated at the scene level. Multiple frame-level estimates are aggregated into a scene-level transform, and the final decision is made by canonical scene acceptance rather than by pairwise usability alone. The pairwise and scene-level tables therefore provide two complementary views of practical registration behavior: correspondence availability and retained scene-level alignment reliability.

Track A evaluates independent pairwise homography evidence, Track B reports the canonical scene-level retain/reject decision, and Track C ranks scenes with the non-trained reliability score for risk–coverage analysis without replacing the canonical gate.

3.5. Label-Free QA Proxies

The protocol relies on label-free relative QA signals designed for scalable scene-level reliability analysis. The strongest stored proxies compare the optimized transform against a baseline initialization and summarize whether scene-level rectification improves cross-modal agreement. The proxy family includes edge-overlap improvement, gradient-NCC improvement, the ratio of frames with improved gradient agreement, severe baseline-relative outlier count and ratio, accepted-frame ratio, and homography-dispersion and robust-rejection diagnostics.

These proxies provide a practical scene-consistency signal for comparing methods and for deciding whether a scene-level transform is reliable enough to retain. They provide scalable scene-consistency diagnostics for reliability-controlled evaluation and can be combined with task-specific annotation protocols when such labels are available, in the spirit of practical matching-oriented evaluation and benchmarking discussed in prior image-matching literature [47, 48].

4. UAV-TAlign Benchmark Reference Framework

4.1. Problem Formulation

To instantiate the scene-level track of UAV-TAlign-12K, we provide UAV-TAlign as its reference alignment framework. For each scene, the input is a set of paired RGB and thermal UAV frames $S = \{(I_i^R, I_i^T)\}_{i=1}^N$, where viewpoint variation, cross-resolution sensing, thermal appearance changes, and day/night conditions make individual frame estimates uneven in quality. The goal is to recover a thermal-to-RGB homography that remains reliable at the scene level, with the canonical output retained when the evidence satisfies the reliability criterion.

This formulation turns registration from a single best-pair problem into a multi-frame reliability optimization problem. In our setting, many scenes already contain usable pairwise correspondences, but the central challenge is to convert uneven frame-level estimates into a transform that is geometrically stable, internally consistent, and acceptable under a reliability-aware criterion. For a selected frame pair, the thermal-to-RGB evidence is represented as a homography $H_i \in \mathbb{R}^{3\times 3}$ acting on homogeneous image coordinates,

$$\tilde{\mathbf{p}}_{ij}^R \sim H_i \tilde{\mathbf{p}}_{ij}^T, \qquad H_i = \arg\min_H \sum_{j \in \mathcal{I}_i} \rho\Big(\big\| \pi(H \tilde{\mathbf{p}}_{ij}^T) - \mathbf{p}_{ij}^R \big\|_2^2 \Big), \tag{1}$$

where $\mathcal{I}_i$ denotes the inlier set selected by robust fitting, $\pi(\cdot)$ converts a homogeneous coordinate to image coordinates, and $\rho(\cdot)$ is the robust estimator loss. The scene-level problem is then to estimate a consensus transform $\bar{H}_S$ from uneven evidence $\{H_i\}$ and decide whether it should be retained as an alignment product.

4.2. Pairwise Evidence Backbone

UAV-TAlign uses a strong off-the-shelf multimodal matcher as its pairwise evidence generator. Our implementation uses the MINIMA family with the RoMa branch as the default backend, while the rest of the pipeline remains agnostic to the exact pairwise matcher [14, 16].

4.3. Reliability-Guided Progressive Evidence Selection

At the scene level, UAV-TAlign starts from a compact, evenly distributed frame subset before expanding the support set. The progressive schedule scales with scene size and can cover all frames when needed. Expansion stops

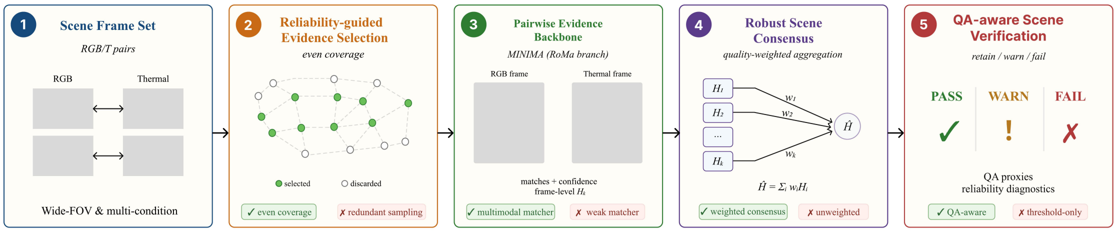

Figure 2: UAV-TAlign benchmark reference framework. Progressive deterministic sampling provides frame-level homography evidence through a MINIMA-RoMa backbone. A fixed geometry-derived quality function weights accepted hypotheses, median/MAD filtering suppresses transform outliers, and QA-aware verification produces the canonical scene decision.

when the accumulated evidence satisfies the accepted-support, consensus-stability, and QA conditions; otherwise, the next deterministic subset is evaluated.

Within each stage, representative frames are selected with deterministic evenly distributed sampling by default. This default policy is used for reproducibility and stable scene coverage. A random-selection variant is reserved for sensitivity analysis and is reported separately.

4.4. Quality-Weighted Frame Geometry

For each selected frame pair, the matcher produces cross-modal correspondences that are filtered into a frame-level homography estimate. UAV-TAlign evaluates each frame using standard geometric statistics, including the number of matches, the number of inliers, inlier ratio, reprojection error, and spatial coverage. A frame enters the global candidate pool only when it passes a geometry-quality gate on these statistics. The overall geometric treatment remains rooted in standard homography estimation and robust fitting practice [25, 24, 26, 27].

Beyond a binary accept/reject decision, the pipeline assigns each accepted frame a quality score so that better-supported homographies receive more influence during scene aggregation. The score is a weighted combination of inlier ratio, spatial coverage, reprojection quality, matcher confidence, and inlier count. This keeps the scene-level estimate from being dominated by low-support frames that only marginally satisfy the frame-level gate. In normalized form, the frame quality weight is written as

$$q_i = \alpha_r \phi(r_i) + \alpha_c \phi(c_i) + \alpha_e \phi(e_i) + \alpha_m \phi(m_i) + \alpha_n \phi(n_i), \qquad \sum_k \alpha_k = 1,\ \alpha_k \geq 0, \tag{2}$$

where r_i is inlier ratio, c_i is spatial coverage, e_i is reprojection quality, m_i is matcher confidence, n_i is inlier support, and $\phi(\cdot)$ maps each diagnostic into a consistent reliability direction. The weights are fixed by the evaluation protocol and are not learned from UAV-TAlign-12K.

4.5. Robust Scene Consensus

Once enough frame-level homographies have been accepted, UAV-TAlign forms a robust scene consensus. Homographies are mapped into parameter space, centered with a median reference, and filtered by a median-absolute-deviation rule to suppress unstable estimates before weighted averaging. The retained homographies are then combined using the frame quality scores defined above. Let $\mathbf{h}_i$ be the scale-normalized vector form of H_i. The consensus set is

$$\begin{aligned}
\mathbf{h}_S^{\text{med}} &= \text{median}_{j \in \mathcal{A}_S}(\mathbf{h}_j), \\
d_i &= \left\| \mathbf{h}_i - \mathbf{h}_S^{\text{med}} \right\|_2, \\
m_d &= \text{median}_{j \in \mathcal{A}_S}(d_j), \\
s_d &= \text{median}_{j \in \mathcal{A}_S} |d_j - m_d|, \\
\mathcal{C}_S &= \{ i \in \mathcal{A}_S : d_i \leq m_d + 2.5 s_d \}.
\end{aligned} \tag{3}$$

where $\mathcal{A}_S$ is the accepted-frame set before robust consensus. The scene transform is then obtained by quality-weighted aggregation,

$$\bar{\mathbf{h}}_S = \frac{\sum_{i \in \mathcal{C}_S} q_i \mathbf{h}_i}{\sum_{i \in \mathcal{C}_S} q_i}, \qquad \bar{H}_S = \text{mat}(\bar{\mathbf{h}}_S). \tag{4}$$

The pipeline records stability diagnostics around the aggregate transform, including homography dispersion relative to the aggregate and the ratio of robustly rejected frame-level estimates. These diagnostics connect consensus formation with the final QA-aware decision.

4.6. QA-Aware Scene Verification

The final decision integrates match availability with scene-level reliability evidence. UAV-TAlign evaluates the optimized scene-level transform against a label-free QA summary computed on representative frames. The QA summary uses cross-modal proxy signals such as edge-overlap improvement, gradient-NCC improvement, the ratio of improved frames, and the count of severe relative outliers. It also retains scene-level match and stability statistics such as accepted-frame ratio, mean inlier ratio, mean reprojection error, spatial coverage, dispersion relative to the aggregate transform, and the robust reject ratio.

The canonical decision uses a baseline-aware verification rule. A scene is retained when accepted-frame statistics remain strong, the aggregated homography is stable, severe baseline-relative outliers remain controlled, and the transform improves gradient-based alignment beyond configured floors. This final verification step gives UAV-TAlign its scene-level character: the framework prioritizes retained scene-level usability over unconditional scene acceptance. For operating-profile visualization, the same diagnostics are summarized by a non-trained reliability score,

$$\begin{aligned} R(S) &= \beta_a A_S + \beta_o(1 - O_S) + \beta_j(1 - J_S) \\ &\quad + \beta_e \Delta E_S + \beta_g \Delta G_S + \beta_\gamma \Gamma_S, \qquad \sum_k \beta_k = 1. \end{aligned} \tag{5}$$

where A_S is accepted-frame support, O_S is severe baseline-relative outlier ratio, J_S is the robust-consensus rejection ratio, ΔE_S and ΔG_S are edge and gradient improvement proxies, and Γ_S collects geometry diagnostics. The strict canonical decision also requires a minimum number of accepted frame hypotheses and one of two fixed QA paths,

$$y_S = \nVdash[N_S \geq N_{\min}] \nVdash[L_S \vee W_S]\,. \tag{6}$$

Here L_S denotes the strict edge/gradient proxy gate and W_S denotes the fixed high-modality path that jointly requires accepted support, stable consensus geometry, controlled relative outliers, and gradient improvement. Thus $R(S)$ ranks scenes for risk-coverage analysis, while y_S defines the retained canonical alignment products reported in the main result.

5. Experimental Setup

5.1. Baselines

We evaluate both classical and learning-based baselines under an explicit off-the-shelf setting. All learning-based methods use public pretrained weights or public pretrained model interfaces and are evaluated without training or adaptation on UAV-TAlign-12K. This choice keeps the comparison focused on practical deployability rather than dataset-specific adaptation. Unified inference wrappers standardize model loading, image I/O, homography fitting, and output schemas while preserving the public pretrained baseline parameters and settings.

The baseline suite covers classical keypoint pipelines, an infrared-visible-specific method, and modern learned matchers. The classical pairwise references are SIFT and AKAZE, each followed by robust homography fitting. RIFT2 is evaluated through its Python implementation under a resize-aware setting. The modern pairwise baselines are Kornia LoFTR, RoMa outdoor full, XoFTR-640, and raw MINIMA. LoFTR uses Kornia's public outdoor-pretrained model, while raw MINIMA uses the MINIMA RoMa branch as a direct pairwise baseline. Reliability-guided progressive evidence selection, robust scene consensus, and QA-aware verification are activated only in the full UAV-TAlign framework [10, 14, 15, 16].

5.2. Shared Geometric Fitting

For the pairwise methods, the common downstream step is homography fitting from the predicted correspondences. After each matcher returns point pairs and optional confidence values, the shared evaluator estimates a homography with a robust backend and then records a unified outcome category. This unified post-processing is important because it makes the pairwise baseline table comparable at the level of usable registration outcomes rather than raw matcher outputs alone.

The shared evaluator uses a robust homography backend across the integrated pairwise baselines:

- robust fitting method: USAC-MAGSAC;
- reprojection threshold: 3.0;
- confidence: 0.999;
- maximum iterations: 10000;
- spatial coverage grid: 4×4.

RIFT2 is independently integrated and uses its frozen evaluated setting: `max_dim=1200`, `npt=2000`, `patch_size=64`, Lowe ratio 0.95, and USAC-MAGSAC with a 5.0-pixel reprojection threshold. Its fitting statistics are therefore reported as method-specific complementary diagnostics rather than treated as a scalar cross-method rank.

5.3. Main-Table Baseline Choices

The main-table baseline choices are as follows. Raw MINIMA uses the MINIMA RoMa backend with the `minima_roma` checkpoint family and the large/full RoMa branch. RoMa uses the public outdoor full model. XoFTR uses the 640-resolution visible-thermal checkpoint family as the default main-table setting. Kornia LoFTR (pretrained="outdoor") uses the public pretrained outdoor model from Kornia.

5.4. Evaluation Details

The LoFTR baseline is evaluated under a resize-aware inference setting. The longest image dimension used for matching is limited to 1200, and automatic mixed precision is enabled. This profile preserves the public pretrained model while supporting consistent full-subset evaluation.

We additionally conduct two controlled protocol validations on the frozen formal outputs. First, five-fold leave-frame-out analysis partitions the accepted frame hypotheses deterministically, recomputes scene consensus after withholding one fold, and measures the resulting grid displacement from the full-evidence transform. Second, a synthetic sensitivity test applies translation, rotation, and scale perturbations at nominal severities of 2, 5, 10, 20, and 40 pixels to three representative frames per scene. It records the response of edge overlap and gradient-NCC without changing the canonical decision rule.

Journal-scale experiments were conducted in isolated Python environments on an NVIDIA RTX 4090 workstation. The main environment used Python 3.10 with PyTorch 2.0.1, torchvision 0.15.2, CUDA 11.8, OpenCV 4.11.0, and Kornia 0.6.11. All runs used the fixed official manifest, fixed public weights or model interfaces, isolated output roots, and read-only raw dataset directories.

6. Results

6.1. Pairwise Infrared-Visible Homography Evidence

Pairwise infrared-visible homography evidence is broadly available on UAV-TAlign-12K. Table 2 reports the full 6,037-pair official evaluation split. Classical keypoint baselines provide useful but less complete pairwise evidence: SIFT and AKAZE reach 87.08% and 94.29% homography availability, respectively. This provides a classical keypoint reference, while modern dense and learned matchers further improve pairwise availability under infrared-visible appearance gaps. Raw MINIMA and RoMa produce homographies for all pairs, while RIFT2 and the strongest learned methods reach complete or near-complete availability. This result sharpens the role of UAV-TAlign: once pairwise homography evidence is broadly available, the main remaining operating question is whether the resulting scene-level product should be retained.

Table 2: Pairwise infrared-visible homography evidence on UAV-TAlign-12K. Metrics are complementary diagnostics under the evaluated implementation and deployment settings, and should not be read as a single absolute ranking. LoFTR uses the declared resize-aware profile, while RIFT2 uses its independently frozen method-specific fitting setting. LoFTR and XoFTR non-ok cases correspond to insufficient matches, no matches, or homography fitting outcomes.

Method	Category	OK / N ↑	H (%) ↑	Inlier Ratio ↑	Coverage ↑	Median Reproj. ↓	Runtime / Pair (s) ↓
SIFT + RANSAC	Classical keypoint	5257/6037	87.08	0.242	0.627	0.000	4.421
AKAZE + RANSAC	Classical keypoint	5692/6037	94.29	0.160	0.730	0.105	1.103
RIFT2 (Python, resize-aware)	Infrared-visible classical	6037/6037	**100.00**	0.086	0.354	0.121	13.980
raw MINIMA	Modern dense / learned	6037/6037	**100.00**	0.319	1.000	1.773	1.346
RoMa outdoor	Modern dense / learned	6037/6037	**100.00**	0.328	0.997	1.741	0.929
Kornia LoFTR outdoor (resize-aware)	Modern dense / learned	5984/6037	99.12	0.074	0.895	0.823	0.169
XoFTR-640	Modern dense / learned	5989/6037	99.20	0.089	0.949	1.732	0.647

The classical baselines remain informative reference points. RIFT2 provides complete infrared-visible homography evidence with low fitted reprojection error, whereas the modern dense and learned matchers provide substantially broader spatial support. These secondary fitting statistics are therefore most informative when read jointly rather than as a single absolute ranking.

6.2. Scene-Level Reference Result

To instantiate the benchmark's scene-level track, we evaluate UAV-TAlign as the reference framework. High pairwise usability does not automatically imply broad scene retention because the canonical decision answers a different operating question. UAV-TAlign produces scene-level homographies for all 15 scenes and retains 9 under the strict canonical reliability gate. This result demonstrates how UAV-TAlign-12K distinguishes homography availability from retained scene reliability.

Table 3: Strict canonical scene-level result for UAV-TAlign on UAV-TAlign-12K. Pair coverage and accepted/attempted support are computed over the nine canonically retained scenes. Arrows indicate the preferred direction.

Method	Eval Unit	H Available ↑	Retained Scenes ↑	Retention Rate (%) ↑	Pair Coverage (%) ↑	Accepted / Attempted	Accepted Ratio (%) ↑
UAV-TAlign	15 scenes	15/15	9/15	60.0	74.4	1491/2304	64.7

6.3. Condition Breakdown

Retained alignments show meaningful condition structure across illumination and thermal rendering. Table 4 separates condition-level scene retention from accepted-frame support aggregated over all scenes within each condition. Day scenes retain most frequently and also provide the strongest accepted-frame support, whereas night scenes are weakest on both axes. Low-light scenes retain less often than day scenes despite a moderate accepted-frame ratio, indicating that frame-level support alone does not determine the final scene-level decision.

Table 4: Scene-level reliability breakdown by light condition on UAV-TAlign-12K. Accepted / attempted aggregates accepted and attempted frames over all scenes within each condition and thus complements the retained-scene counts.

Condition	Scenes	Retained ↑	Retention (%) ↑	Accepted / Attempted ↑	Accepted Ratio (%) ↑
Day	7	**5**	**71.4**	**1650/2203**	**74.9**
Night	5	3	60.0	361/890	40.6
Low-light	3	1	33.3	462/754	61.3

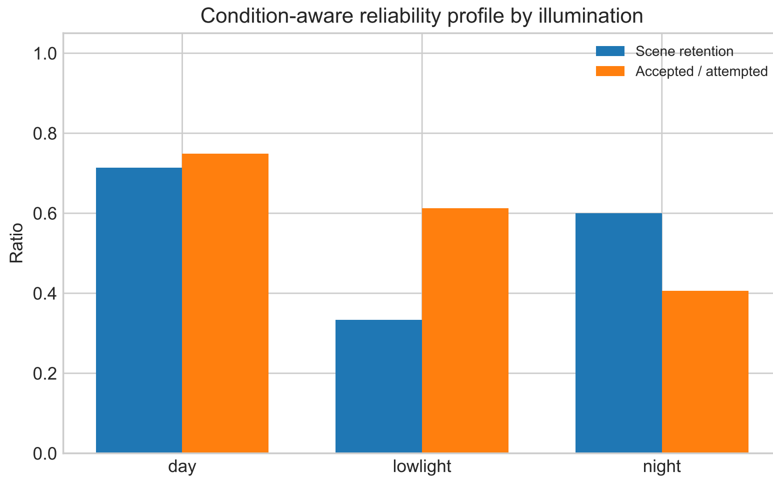

Figure 3: Condition-aware reliability profile by illumination. Scene retention and accepted-frame support summarize complementary aspects of the retained alignment products.

6.4. Reliability-Coverage Operating Profile

The canonical QA gate defines retained alignment products, while the reliability score is used for ranking, threshold sweep, and operating-profile visualization. The score-ranked top-K subsets and the canonical retained-scene set are related but not identical. Figure 4 therefore marks the strict canonical $9/15$ operating point explicitly while plotting the score-ranked reliability–coverage curve.

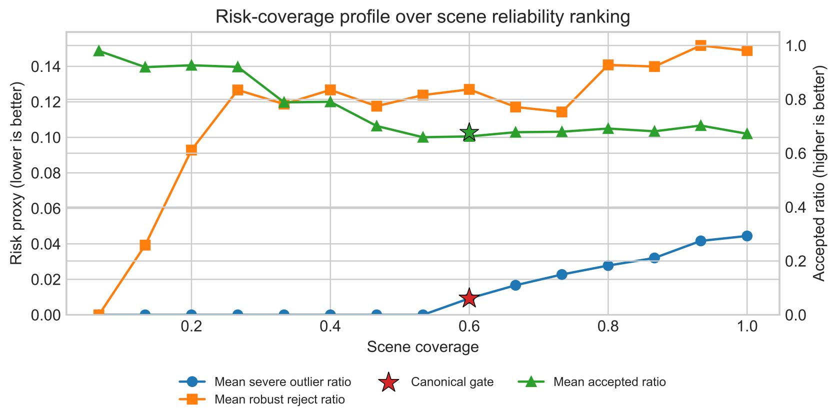

Figure 4: Risk-coverage operating profile on UAV-TAlign-12K. The curves are generated by ranking scenes with the non-trained reliability score and progressively increasing retained coverage. The star marks the strict canonical QA-gate operating point (9/15 retained scenes). The score-ranked top-K operating points visualize the reliability–coverage profile and do not replace the canonical decision rule.

At the canonical point, UAV-TAlign retains 9 scenes, covers 60.0% of scenes and 74.4% of evaluated pairs, and keeps the mean severe-outlier ratio at 0.009. The score-ranked curve exposes a smooth operating profile from conservative high-confidence retention to broader inclusive coverage. We keep the canonical gate fixed and use this curve to visualize the reliability–coverage trade-off.

6.5. Controlled Protocol Validation

Controlled resampling and geometric perturbations independently validate the behavior measured by the scene-level protocol. The five-fold leave-frame-out test covers all 15 scenes and 75 consensus recomputations. With approximately one fifth of accepted frame hypotheses withheld per fold, the scene-median displacement from the full-evidence consensus is 1.39 pixels (mean 2.26 pixels).

The perturbation test covers 45 representative frames and 675 controlled trials. Within each perturbation family, increasing severity produces a monotonic decrease in the joint edge/gradient quality signal. At the largest tested severity, the mean joint-score changes are -0.074, -0.031, and -0.025 for translation, rotation, and scale, respectively; 84.4%, 80.0%, and 73.3% of the corresponding trials show degradation. Table 5 summarizes the independent evidence.

Table 5: Controlled validation of scene-consensus stability and QA sensitivity. Synthetic degradation rates and joint-score changes are reported at the largest tested severity.

Validation	Scope	Primary response	Degraded trials (%)
Leave-frame-out consensus	15 scenes / 75 folds	1.39 px scene-median drift	–
Synthetic translation	45 frames / 225 trials	ΔJoint = −0.074	84.4
Synthetic rotation	45 frames / 225 trials	ΔJoint = −0.031	80.0
Synthetic scale	45 frames / 225 trials	ΔJoint = −0.025	73.3

6.6. Cumulative Ablation

The fixed 8-scene supplement separates the contribution of robust consensus from final scene verification. Relative to pairwise evidence alone, robust consensus raises mean edge gain from 0.124 to 0.146 and mean gradient-NCC gain from 0.126 to 0.149, while reducing severe-outlier ratio from 0.104 to 0.042 and mean transform dispersion from 151.2 to 52.5 pixels. QA-aware verification then converts the permissive stage output into the final retained-product decision.

Table 6: Frozen cumulative ablation on the fixed 8-scene subset. The table emphasizes retained product behavior under the accepted local supplement evidence.

Variant	Selection	Consensus	Verification	Stage Pass/N ↑	ΔEdge ↑	ΔGrad ↑	Severe ↓	Disp. (px) ↓
Pairwise evidence only	even	–	–	8/8	0.124	0.126	0.104	151.2
+ Robust scene consensus	even	✓	–	8/8	0.146	0.149	0.042	52.5
+ QA-aware verification	even	✓	✓	5/8	0.146	0.148	0.042	57.7

The accepted random-selection supplement further shows a 5/8 to 7/8 spread across seeds on the same fixed subset. The variation is concentrated in a small number of borderline scenes, while the accepted-frame means remain comparatively stable across seeds. Deterministic-even selection therefore serves as a reproducible default operating policy, and the multi-seed result is best read as boundary sensitivity rather than full-pipeline instability.

7. Discussion

7.1. What UAV-TAlign-12K Reveals

The main contribution of UAV-TAlign-12K is a captured dataset and protocol that expose a gap hidden by pairwise-only reporting. Modern matchers provide abundant frame-level evidence on the official split, yet the strict scene-level criterion retains only a subset of scenes. The benchmark therefore separates two practically different capabilities: producing a pairwise homography and deciding whether multi-frame evidence supports a trustworthy scene transform.

The protocol evidence supports this dataset-centered reliability framing. Retained and QA-filtered scenes show structured differences in accepted ratio, severe outlier behavior, and baseline-relative QA signals, indicating that the canonical criterion captures scene-level reliability beyond pairwise matcher availability alone. This gives the evaluation a clear diagnostic role: it tests when abundant pairwise evidence becomes a trustworthy scene-level transform. The canonical scene criterion is therefore designed for conservative retention of scene transforms rather than for maximizing unconditional scene acceptance.

The controlled validations reinforce this interpretation from two independent directions. Leave-frame-out recomputation keeps the scene consensus close to its full-evidence estimate, while known translation, rotation, and scale perturbations progressively reduce the joint QA signal. Together, these results connect multi-frame consensus stability with sensitivity to controlled geometric degradation.

UAV-TAlign serves as a reference implementation for this second evaluation unit. Its evidence selection, robust consensus, and QA-aware verification demonstrate one reproducible way to convert pairwise hypotheses into a retained scene product; the benchmark remains compatible with future matchers and alternative scene-level aggregation strategies.

The multi-seed random-selection supplement further strengthens the design rationale for deterministic evidence selection. On the accepted 8-scene subset, random selection spans 5/8 to 7/8 retained scenes across seeds, but the variation is concentrated in a few borderline scenes rather than in large shifts of accepted-frame support. Deterministic-even selection therefore provides a reproducible default operating policy, while robust scene consensus and QA-aware verification define the retained-product decision layer.

7.2. Operating Scope and Extensibility

UAV-TAlign-12K targets captured UAV scenes where a dominant planar or far-field transform is a useful operating model. Its 15 scenes span complementary illumination, thermal rendering, view, and scene-family conditions, establishing a focused evaluation resource for cross-FOV homography alignment. The same manifest and evaluation interfaces can be extended to additional flight altitudes, sensor configurations, and geometric models.

The label-free QA proxies provide scalable scene-consistency diagnostics, while the fixed canonical gate and the non-trained reliability ranking expose complementary retained-product and operating- profile views. This modular design supports direct combination with task-specific annotations, alternative scene geometries, learned quality estimators, and downstream thermal-sensing tasks.

8. Conclusion

This work introduces UAV-TAlign-12K as a real-world cross-FOV UAV infrared-visible dataset and scene-level reliability benchmark. Its 6,039 pairs, integrity-checked official split, diverse capture conditions, and manifest-fixed protocol support reproducible evaluation at both the pairwise and scene levels. The benchmark shows that broad homography availability does not by itself guarantee a scene transform that should be retained.

UAV-TAlign provides the benchmark's reproducible scene-level reference framework based on reliability-guided progressive selection, robust consensus, and QA-aware verification. It produces scene homographies for all 15 scenes and retains 9 under the strict canonical gate. Controlled leave-frame-out and perturbation tests further verify consensus stability and QA sensitivity. Together, the dataset, protocol, and reference framework provide a practical foundation for developing and evaluating more reliable UAV infrared-visible alignment systems.

Data Availability

UAV-TAlign-12K, the official evaluation manifest, metadata, integrity report, and evaluation scripts have been prepared for public release. The official evaluation split contains 6,037 integrity-checked RGB-infrared pairs from a 6,039-pair UAV infrared-visible alignment collection. The dataset will be released under the Creative Commons Attribution 4.0 International license, with code and evaluation materials provided through a public repository. Review access can be provided during peer review. The official manifest hash is provided for reproducible evaluation.

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